Leading asymptotic of the chart standard integral: $\int_0^b u^h e^{-\beta n u^{2k} + \beta\sqrt{n} u^k a}\, du \sim \frac{1}{2k} S_{(h+1)/(2k)}(a)\, n^{-(h+1)/(2k)}$

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2026-09-06

Abstract

The leading term of `cor:standardintegralexp` in one dimension, stated as a genuine `Asymptotics.IsEquivalent`: as $n \to \infty$ the population standard integral over the chart $(0, b]$ is asymptotically $\frac{1}{2k} S_{(h+1)/(2k)}(a)\, n^{-(h+1)/(2k)}$. The exponent $\mu_1 = (h+1)/(2k)$ — the first element of $\Lambda(h, k)$ — is sharp because the fluctuation function is strictly positive (`fluctuation_pos`, also new). Zero `sorry`/`axiom`.

1 Setting

Units 22 and 25 give the chart integral as the half-line value $\frac{1}{2k} n^{-\mu_1} S_{\mu_1}(a)$ plus an error bounded by $K e^{-\varepsilon n}$, $\varepsilon = \frac{\beta}{4} b^{2k}$. Since $e^{-\varepsilon n} = o(n^{-\mu_1})$ and the coefficient is nonzero, this is exactly the statement $u \sim v$ in Mathlib's sense ($u - v = o(v)$). The unit packages that observation in the seabed's standard `IsEquivalent` form (cf. the Wong-arc and `TwoD` asymptotic packaging).

2 The things formalised

```
theorem fluctuation_pos (  lam a :  ) (h  : 0 <  ) (hlam : 0 < lam) :
    0 < fluctuation   lam a

theorem standardIntegral1D_chart_isEquivalent (  a b :  ) (h k :  )
    (h  : 0 <  ) (hb : 0 < b) (hk : 0 < k) :
    (fun n :   =>   u in Ioc 0 b, u^h * exp(- *n*u^(2*k) +  *√n*u^k*a))
      ~[atTop] fun n :   => (1/(2*(k: ))) * n^(-(((h: )+1)/(2*k)))
        * fluctuation   (((h: )+1)/(2*k)) a
```

3 Proof strategy

`IsLittleO.isEquivalent` reduces the claim to $u - v = o(v)$. First $u - v = O(e^{-\varepsilon n})$ by `IsBigO.of_bound` with the constant $K = e^{\beta a^2/2} (\beta/4)^{-\mu_1} (2k)^{-1} \Gamma(\mu_1)$ from `standardIntegral1D_bounded_approx`, eventually for $n \geq 1$. Then `isLittleO_exp_neg_mul_rpow_atTop` gives $e^{-\varepsilon n} = o(n^{-\mu_1})$, and `IsLittleO.const_mul_right` with the nonzero constant $\frac{1}{2k} S_{\mu_1}(a)$ turns the right-hand side into v; `IsBigO.trans_isLittleO` finishes. Positivity of $S_\lambda(a)$ is `setIntegral_pos_iff_support_of_nonneg_ae`: the integrand is positive on $(0, \infty)$, so its support meets $(0, \infty)$ in the whole half-line, of infinite measure.

4 Roadblocks and resolutions

None of substance: every step compiled on the first attempt. Two idioms recorded: the exponent match $-\frac{\beta}{4}nb^{2k} = -\varepsilon n$ is a `show …by ring` rewrite inside the calc (the exponential is opaque to `ring`); and `Real.volume_Ioi` plus `ENNReal.zero_lt_top` supply the positive-measure input to the support criterion.

5 What was Mathlib, what was new

Mathlib: `IsLittleO.isEquivalent`, `IsBigO.of_bound`, `IsBigO.trans_isLittleO`, `isLittleO_exp_neg_mul_rpow_atTop`, `IsLittleO.const_mul_right`, `setIntegral_pos_iff_support_of_nonneg_ae`. New: strict positivity of the fluctuation function and the packaged leading asymptotic with its sharp exponent.

6 Lessons

A closed-form-plus-exponential-tail identity is one lemma away from an `IsEquivalent`; state the asymptotic as soon as the tail bound lands, since that is the form downstream consumers (and the paper's "$\sim$") actually use.

7 Follow-ups

The full one-dimensional expansion to all orders (all of $\Lambda(h, k)$ with $\log n$-free polynomial coefficients, since $d = 1$) for a monomial perturbation is now a summability/tail-bookkeeping exercise on `standardIntegral1D_monomial_expansion`; the multivariate tree remains the substantive gap.